THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which of the following is a rational number? 1

(a) $\sqrt{5}$ **(b)** π **(c)** $\frac{7}{9}$ **(d)** $\sqrt{7}$

2. $(x+2)(x+3) =$ 1

(a) x^2+x+6 **(b)** x^2+6x+6 **(c)** x^2+5x-6 **(d)** x^2+5x+6

3. The point $(-4, 2)$ lies in: 1

(a) Quadrant IV **(b)** Quadrant I **(c)** Quadrant II **(d)** Quadrant III

4. The degree of the polynomial x^3-2x+1 is: 1

(a) 2 **(b)** 5 **(c)** 4 **(d)** 3

5. Which of the following is an irrational number? 1

(a) $\sqrt{25}$ **(b)** $\frac{5}{7}$ **(c)** $\sqrt{3}$ **(d)** 5.5

6. $(x+3)^2 =$ 1

(a) x^2+6x+9 **(b)** x^2+9 **(c)** x^2-6x+9 **(d)** x^2+3x+9

7. The abscissa of the point $(-7, 3)$ is: 1

(a) -8 **(b)** -5 **(c)** -6 **(d)** -7

8. If $p(x) = 5x-1$, then $p(3) =$ 1

(a) 14 **(b)** 13 **(c)** 15 **(d)** 16

9. The decimal expansion of $\frac{7}{8}$ is: 1

(a) 1 **(b)** 0.875 **(c)** 8.75 **(d)** 0.0875

10. $(x-5)^2 =$ 1

(a) x^2+25 **(b)** $x^2-10x+25$ **(c)** $x^2-5x+25$ **(d)** $x^2+10x+25$

11. The ordinate of the point $(-3, 7)$ is: 1

(a) 7 **(b)** 9 **(c)** 8 **(d)** 6

12. The zero of the polynomial $2x-6$ is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 5

13. The decimal expansion of $\frac{2}{3}$ is: 1

(a) $0.\overline{6}$ **(b)** 1 **(c)** $0.0\overline{6}$ **(d)** $6.\overline{6}$

14. $(x+4)(x-4) =$ 1

(a) $x^2+8x-16$ **(b)** x^2+16 **(c)** $x^2-8x+16$ **(d)** x^2-16

15. The distance of the point $(5, 12)$ from the origin is: 1

(a) 13 **(b)** 12 **(c)** 14 **(d)** 15

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Evaluate: $(-9) \times (-4)$. 2

17. Find 42^2 using $(a+b)^2$. 2

18. Write the coordinates of the point that lies 5 units left of the y-axis and 2 units above the x-axis. 2

19. If $p(x) = 4x - 3$, find $p(5)$. 2

20. Write $\frac{5}{8}$ as a decimal. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Express $0.\overline{4}$ in the form $\frac{p}{q}$. 3

22. Expand $(3x+1)^2$. 3

23. Write the quadrant of $(-4, 6)$, and the reflections of this point in the x-axis and in the y-axis. 3

24. Complete the table of values for $y = 2x + 5$ at $x = 0, 1, 2$. 3

OR

Express $\frac{7}{15}$ as a decimal and say whether it terminates or repeats. 3

25. Factorise $x^2 + 8x + 16$. 3

OR

Find the midpoint of $A(1, 1)$ and $B(7, 3)$ and its distance from the origin. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The vertices of a triangle are $A(1, 2)$, $B(11, 2)$ and $C(11, 26)$. Find the three sides and show that the triangle is right-angled at B. 5

OR

A taxi charges a fixed fare of ₹20 plus ₹12 per km. Write the fare $F(x)$ for x km, find the fare for 5 km and the distance covered for a fare of ₹104. 5

27. Classify each of $\sqrt{2}$, $\sqrt{49}$, $\sqrt{64}$, $\frac{5}{9}$, 0.45 as rational or irrational. 5

OR

Expand $(3x + 2y)^2$ and find its value when $x = 1$ and $y = 1$. 5

END OF QUESTION PAPER — **Total: 50 Marks**

MASTERMIND